

Combinatorics – 3

Combinatorial numbers: Fibonacci, Catalan, partitions

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Theoretical minimum. Two families of numbers carry almost all of the olympiad load, and both are worth knowing by heart together with their identities.

Definitions. $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$, and $L_1 = 1$, $L_2 = 3$, $L_{n+2} = L_{n+1} + L_n$ (Lucas). The Catalan numbers are $C_0 = 1$, $C_{n+1} = \sum_{k=0}^n C_k C_{n-k}$. The number of partitions of n is $p(n)$, the Stirling number of the second kind $S(n, k)$ counts the partitions of an n -set into k non-empty blocks, the Stirling number of the first kind $c(n, k)$ counts the permutations of $[n]$ with k cycles, and the Bell number $B_n = \sum_k S(n, k)$. Closed forms for F_n and C_n via generating functions are in Functional sequences, series and spaces – 3, homework 9, and are used here without repetition.

Theorem 1 (the Fibonacci toolbox). Cassini: $F_{n-1}F_{n+1} - F_n^2 = (-1)^n$; the addition formula $F_{m+n} = F_m F_{n+1} + F_{m-1} F_n$; the matrix form $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, from which both of the above follow by taking determinants and by multiplying; divisibility $\gcd(F_m, F_n) = F_{\gcd(m, n)}$; and the identity $\sum_k \binom{n-k}{k} = F_{n+1}$, which is the tiling interpretation. Every positive integer has a unique *Zeckendorf representation* as a sum of non-consecutive Fibonacci numbers.

Theorem 2 (Catalan in five guises). C_n counts: the Dyck paths of length $2n$; the balanced bracket sequences with n pairs; the triangulations of a convex $(n+2)$ -gon; the binary trees with n internal nodes; and the ballot sequences in which one candidate is never behind. All five are linked by explicit bijections, and the cycle lemma gives $C_n = \frac{1}{n+1} \binom{2n}{n}$ directly.

Fact 1 (partitions). $\sum_n p(n)x^n = \prod_{k \geq 1} (1 - x^k)^{-1}$; conjugating a Young

diagram shows that the partitions into at most k parts are as many as those into parts of size at most k ; Euler's pentagonal number theorem $\prod (1 - x^k) = \sum_j (-1)^j x^{j(3j-1)/2}$ turns into a fast recurrence for $p(n)$.

Fact 2 (binomial arithmetic). Vandermonde $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$; Lucas' theorem computes $\binom{n}{k}$ modulo a prime p from the base- p digits, whence the number of odd entries in row n of Pascal's triangle is $2^{s(n)}$, where $s(n)$ is the number of ones in binary.

1. Prove the matrix form of the Fibonacci recurrence and deduce Cassini's identity, the addition formula and $\gcd(F_m, F_n) = F_{\gcd(m,n)}$. Then prove the existence and uniqueness of the Zeckendorf representation.
2. Let a_n be the number of sets S of positive integers for which $\sum_{k \in S} F_k = n$. Determine a_n , or describe the values it takes. (Putnam 2020)
3. Let $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$. For $m > 2$ let R_m be the remainder when the product $\prod_{k=1}^{F_m-1} k^k$ is divided by F_m . Prove that R_m is again a Fibonacci number. (Putnam 2021)
4. Prove the equivalence of the five descriptions of the Catalan numbers by explicit bijections, and prove $C_n = \frac{1}{n+1} \binom{2n}{n}$ by the cycle lemma. Then solve the ballot problem: in an election with $2n$ valid votes, what is the probability that one candidate is never behind? (ISTCiM 2021)
5. Let n be a positive integer. Prove that 2^{n-1} divides $\sum_{0 \leq k < n/2} \binom{n}{2k+1} 5^k$, and explain why the Fibonacci numbers are hiding in this sum. (IMC)